Review
On-line monitoring of wastewater quality: a review

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Abstract: Real-time monitoring of wastewater quality remains an unresolved problem to the wastewater treatment industry. In order to comply with increasingly stringent environmental regulations, plant operators as well as instrument manufacturers have expressed the need for new standards and improved comparability and reliability of existing techniques. A review of currently available methods for monitoring global organic parameters (BOD, COD, TOC) is given. The study reviews both existing standard techniques and new innovative technologies with the focus on the sensors' potential for on-line and real-time monitoring and control. Current developments of biosensors, optical sensors and sensor arrays as well as virtual sensors for the monitoring of wastewater organic load are presented and the interests and limitations of these techniques with respect to their application to the wastewater monitoring are discussed.

Keywords: BOD; COD; TOC; on-line instrumentation; real-time monitoring; sensors; wastewater

1 INTRODUCTION

Monitoring of wastewater quality parameters is currently a subject of growing concern both in the United Kingdom and internationally. Indeed, a number of European regulatory measures and recommendations, such as the 91-271 EEC directive, have put pressure on the water and wastewater treatment industries with respect to discharge requirements. Traditionally, the quality of treated wastewater is defined by the measurement of global parameters such as Biological Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Organic Carbon (TOC), and Total Suspended Solids (TSS).[1–3] In addition to providing vital information on the quality of treated wastewater and treatment efficiency, these procedures demonstrate that a wastewater treatment plant meets statutory laws.

In order to comply with these regulations on a permanent basis, and because of the spatial and time dependent variability of wastewater characteristics, on-line monitoring of the above parameters is clearly needed. At present, however, there is often infrequent monitoring of wastewater quality at most treatment plants. The automation of wastewater systems is not as developed as other process industries, mainly because of the hostile environment in which sensors have to be located.[4] The lack of sensors suitable for on-line real-time monitoring/control is often due to uncertainties regarding the reproducibility/reliability of existing methods, whereas more sensitive and standardised laboratory-based techniques are time consuming, and require sample collection and retrospective analysis. Furthermore, straightforward extrapolation of laboratory measurements is not satisfactory as a grab sample (taken on a daily basis in the best case scenario) is unlikely to provide a meaningful and high resolution picture of the nature of, and variation in, wastewater quality.

In Europe, the owners and operators of treatment plants together with the producers of monitoring equipment have expressed an urgent need for new standards and the improvement of comparability, reliability and quality of existing techniques, as well as to develop new, fast-responding technologies.[5,6] In this perspective, and with the growing need for on-line, and real-time, monitoring of wastewater quality, it became necessary to present and discuss the state of the art of wastewater monitoring techniques for global parameters.

This present study reviews the existing standard and most widely applied methods for organic load monitoring (BOD, COD, TOC). Less common but novel and potentially useful techniques are also presented with a focus on on-line, and real-time, monitoring and control.

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2 PRINCIPLES AND CLASSIFICATION OF EXISTING TECHNIQUES

In addition to traditional laboratory-based analytical techniques used in the water industry, recent years have seen the development of a range of innovative monitoring equipment. Although only a small number of such product has yet reached the market or has been accepted, there is already a great diversity of techniques and technologies available, both commerically and in research laboratories, which are reported in the literature. As a consequence, different schemes have been used in an attempt to classify existing sensors and analysers according to their respective properties.

Lynggaard-Jensen[4] listed eight different sensor/analyser properties (Table 1) which should be taken into consideration before their introduction into wastewater systems (ie for monitoring or process control). Indeed key features such as the cost of ownership, ease of use, placement of the sensors, as well as the response time, will influence the consumer's choice. Other technical aspects such as the principle of measurement, reliability, accuracy and detection limits will also dictate whether or not the technology will be accepted and promoted as a standard (or alternative) method by the end user and relevant authorities. It is, therefore, evident that both the performance characteristics (range, linearity, accuracy, response time, limit of detection, etc.) and the intrinsic properties of the sensors (single or multi-parameter, need for external sampling and filtration, intrusive/non-intrusive) are of major importance when looking at existing and new methodologies for wastewater systems.

3 MONITORING ORGANIC POLLUTION IN WASTEWATER: STANDARD METHODS

New sensors/analysers reported in the literature are discussed. It is often the case that while sensors are developed to measure parameters for other applications, they can be adapted to the wastewater industry.

Gross discharges of untreated or insufficiently treated wastewater effluent into receiving waters are generally considered to be one of the most important and common forms of water pollution.[7] Because of the complexity and variability of wastewater quality, 'Blanket' determinations, which measure a range of similar substances rather than individual compounds, are usually carried out. When considering organic substances in wastewater, it is customary to measure the amount of oxygen required to oxidise these substances, this then provides a measure of the strength or polluting potential of the sample.[8–11]

3.1 Biological oxygen demand

Traditionally, organic pollution is measured by the standard, off-line, 5-day Biological Oxygen Demand test (BOD_5), mainly as proof of compliance with relevant legislation.[8] This test is an empirical test indicator of biological activity. It is defined as the potential for removal of oxygen from water by aerobic heterotrophic bacteria which utilise organic matter for their metabolism and reproduction.[13] In practice it is essentially a measure of the amount of dissolved oxygen required for the biochemical oxidation of organic compounds in 5 days. This gives an indication of the waste biodegradability and is, therefore, a desirable measurement in biological treatment processes.[8,14,15]

The standardised test was first published in *Standard Methods* in 1917.[16] There are many problems associated with the application of the test and the meaning of the results which are well documented, and its limitations have been extensively reviewed.[17–19] Although the test has been refined over the years, the basic approach of using a dilution technique has remained essentially unchanged.[20] Unfortunately these dilutions reduce the concentration of substrates and micro-organisms in the samples, thereby decreasing overall kinetic rates.[20] This, in conjunction with the arbitrary time period of 5 days, may not reflect the conditions in the treatment process[20] and must be taken into account when interpreting the results.

Another problem with the test lies in the presence of toxic substances in wastewater which can affect (or even kill) the bacteria, even at low concentrations. For instance, heavy metals such as lead, copper, mercury or chromium have been shown to inhibit, or completely prevent, the oxidation of organic waste by bacteria.[21] The extent to which toxic substances affect BOD values is well documented.[22–24] As, in practice, such substances are rarely monitored, it is often difficult to tell if a decrease in BOD values is only due to a decrease in the organic load. Although the BOD test still provides the best estimate of the reactivity of the contaminants in the natural environ-

Table 1. Relevant sensor properties (after Lynggaard-Jensen[4])

Property	*Example*
1 Placement of sensor	*In-situ*, at-line, in-line; on-line, off-line
2 Principle of sampling	External sampling, no external sampling
3 Principle of filtration	Filtration, no filtration
4 Principle of sample treatment	Continuous, batch
5 Principle of measurement	Photometric, colorimetric, enzymatic, titrimetric, etc
6 No. measurands	Single parameter, multi parameter
7 Need for supplies	Consumables, no consumables
8 Service intervals	Long, medium or short intervals

ment, it remains insensitive and imprecise at low concentrations.[10] Last, but not least, the BOD test is slow to yield information, labour intensive, and, because of the dilutions and other manipulations, takes a long time to complete. Since it takes 5 days from sample collection to the acquisition of the results, the conventional method is not suitable for process control and real-time monitoring where rapid feedback is essential. In the case of a pollution event or an operation problem, adverse conditions would be unknown and perhaps persist for 5 days until the outcome of the BOD test is known.[20]

Despite all these limitations, the 5-day BOD test is still extensively used and preferred over other tests.[8,25] Logan and Patnaik[26] noted the crude character of the standard method, when compared with modern analytical techniques. Yet a substantial amount of time and money is still devoted to measuring BOD at wastewater treatment plants. The only recent improvement involved measuring the Dissolved Oxygen (DO) using a DO probe which often requires daily calibration (Winkler method) to ensure probe accuracy. Finally, the method requires experience and skills, and it generally has an uncertainty of 15–20% in the results. It is argued that the heterogeneity of microbial populations in wastewater treatment plant systems and their widely differing responses to substrates, in combination with the variability of wastewater compositions could be responsible for these discrepancies.[27]

3.2 Chemical oxygen demand

In an attempt to overcome the difficulties in achieving reproducible results, as well as speed and precision, a series of wet chemical techniques have been developed.[7] The use of COD and TOC as analytical parameters to monitor organic pollution has become more common in recent years.

The COD test is now widely used as a means of measuring the organic strength of domestic and industrial waste, often replacing BOD as the primary parameter in wastewater. It is based upon the fact that most organic compounds can be oxidised by the action of strong oxidising agents under acid conditions.[28] As for BOD, it essentially consists of measuring the amount of oxygen required. In the case of COD however, organic matter is converted to carbon dioxide and water regardless of the biological assimilability of the substances. For example, glucose and lignin are both oxidised completely.[28]

The major advantage of the COD test is that the results can be obtained within a relatively short time (approx. 2h instead of 5 days for the BOD_5). Additionally, it was shown that the presence of toxic substances would not affect COD measurements.[22] In many cases a linear relationship between COD and BOD values can be established,[21] allowing COD data to be interpreted in terms of BOD values. However, this relationship may be time dependent and is likely to be affected by changes in wastewater quality. Such prediction requires the use of a solid experience-based model and reliable correlation factors,[28] with particulars are being given UNCLAR TEXT sudden changes in wastewater composition. When used in conjunction with BOD, the COD test can provide an indication of the biodegradability of the wastewater (the BOD/COD ratio is frequently used) and can also be helpful in the detection of toxic conditions.

One of the main limitations of the COD test is its inability to differentiate between biodegradable and biologically inert organic matter on its own. Therefore, its use as a parameter in controlling biological treatment plants remains un question. In addition, the use of chemicals such as acid, chromium, silver and mercury produce liquid hazardous waste which requires disposal. This has largely been responsible for the lack of uptake by the water industry. Finally, limited precision and accuracy of the test at values less than $5\,mg\,dm^{-3}$ has been reported by APHA and others.[29]

3.3 Total organic carbon

Since its appearance in the 1970s, the use of TOC as an analytical parameter has become more common. This is particularly true in the case of industrial wastewater where it is often considered as the most relevant or 'true' parameter for the global determination of organic pollution.[30] One of the major advantages lies in the rapidity of the test. Determination can be carried out in triplicate within minutes, allowing a much greater number of measurements than is possible with BOD or COD tests.

Two main techniques are usually used for the conversion of organic carbon to carbon dioxide for TOC determination.[30] In the first one, which is sometimes called Wet Chemical Oxidation (WCO), oxidation is performed at low temperature by ultraviolet light and the addition of persulfate reagent, after removal of inorganic carbon by acidification and aeration. The second, and more recent method uses a catalyst at high temperature (650–900 °C) and is known as HTCO (High Temperature Catalytic Oxidation). Recent studies have, however, shown significant differences and conflicting results between the two techniques.[30–32] As a result, both methods are still being investigated and their accuracy is still subject to controversy. (Statham, personal communication).

In addition, there is still considerable resistance to its use for municipal wastewater because of the difficulty faced in obtaining good TOC–BOD_5 correlations.[33,34] This is particularly important in studies on the kinetics of wastewater treatment processes.[34] In fact, TOC only measures the content of organic compounds, not other substances that may contribute to BOD.[35]

Given these considerations, it is clear that the standard techniques of monitoring wastewater BOD, COD and TOC all pose some problems to the end user as well as the legislator. Amongst the most important limitations are the time required (5 days in the case of BOD_5), the lack of reliability and the difficulty in achieving reproducible results. Although both COD and TOC are more effective in terms of speed and accuracy, they are unable to differentiate between biodegradable and non-biodegradable matter. Furthermore, these methods are mainly based on sample collection and retrospective analysis. In order to comply with the regulations and using standards for environmental protection, wastewater treatment plants need to implement automated measuring techniques. Sensors and other analytical tests in continuous or sequential mode would be suitable for alarm systems, and would facilitate process control and plant operation strategy.

Following the increasing demand for real-time monitoring, a considerable amount of literature has been published. Several innovative techniques and prototypes have been developed on a laboratory-scale in some universities, and some instrument manufacturers are now offering devices that perform rapid monitoring of wastewater organic strength. However, there is still a gap between the research and development of new techniques, and their effective implementation by the end user. The water industry remains slow in taking up new technologies because of the lack of recognised and standardised methods or instruments that would satisfy all their practical requirements. Consequently, in order to improve the comparability, reliability and quality of measurements obtained from *in-situ* on-line sensors/analysers, new projects such as the ETACS (European Testing and Assessment of Comparability of on-line Sensors/analysers) project have recently been initiated (project 3256 ETACS).[36] This project aims to achieve standardised validation of methods for *in-situ* on-line measurement of water quality determinants such as BOD, COD and TOC[37] which should help fill the wide gap that currently exists between laboratory or field research results of sensor technology and their implementation by the end user.

4 ALTERNATIVE NEW TECHNIQUES

A review of the techniques currently available is now presented with emphasis on automatic on-line methods for real-time monitoring.

There is a plethora of new techniques and technologies available for monitoring changes in organic load in wastewater. The purpose of these is to obtain specific information about changes in wastewater quality (particularly at the inlet) and to ensure treatment efficiency for compliance assessment and process control. Accordingly, several instrument manufacturers are now offering devices to perform rapid monitoring of wastewater BOD, but little experience with the technology has yet accumulated.[38] Here is a selection of more or less developed systems described in the literature (Tables 2, 3 and 4).

In order to meet the requirements for real-time and on-line monitoring in the real world, all methods must be simple, fast and reliable, as well as cost effective (reagents, maintenance, energy, operator's time, etc). With this in mind, several studies have been carried out to test and compare various types of analysers.[12,39,40]

Brookman[13] noted the wide range of existing methods for the determination of BOD. These include the use of a microbial sensor;[41] potentiometric stripping analysis;[42] incubation in acidified N/80 permanganate for 4 h to give the permanganate value;[43] and the use of a scanning optical sensor based on the oxygen quenching of luminescence.[44]

Accordingly, it is possible to distinguish between a few major types of techniques: biosensors and chemical, optical, or even 'virtual' (software, modelling) sensors.

4.1 Biosensors

In the field of biosensing, respirometric techniques have been a popular method of measuring biodegradable organics and a detailed description of different systems can be found in the *IAWQ Scientific and Technical Report No 7*.[45]

Although some techniques use free cells in solution,[46] the basic structure of microbial (BOD) sensors generally consists of an oxygen probe and immobilised micro-organisms. Tan and Qian[47] reported that the microbial system used in the preparation of (BOD) biosensors usually consists of a single type of micro-organism with species varying from one system to another. Similarly, the medium and the methods of immobilisation are generally different, and membranes of different types and pore sizes are used.[9,38,48] Biosensors based on immobilised bacteria are now starting to be commercially available for operational use.[4] An example of such a biosensor is shown in Fig 1.

Some of the major problems currently encountered when using biosensors for monitoring BOD have been discussed by Iranpour *et al*,[27] Tan and Qian[47] and Qian and Tan.[48] In particular, they pointed out the risks associated with the use of infectious (or non-infectious) microbes, both for the environment and for the users.

Additionally, biosensors that use single species of micro-organisms, and of which the measurements are essentially based on the correlation between the concentration gradient of Dissolved Oxygen (DO) across the biofilm membrane composite and the BOD_5 equivalence of glucose–glutonic acid (GGA) BOD check solutions (according to Japanese Industrial Standards, 1993), are unlikely to be representative of the conditions that prevail in wastewater treatment plants (Fig 2). This is in contrast with the conventional BOD_5 test. However, biosensors offer the possibility to measure sterilised and synthetic samples which usually require seeding and acclimation of the micro-organisms. Thus, Tan and Qian[47] and Qian and Tan[48] showed the advantage of thermally killed and multi-species systems (now commercially available) and

Table 2. Current techniques in monitoring wastewater organic load: biosensors

Parameter	Description	Applications	Range and Response time	Interests	Limitations	Reference
BOD	Yeast impregnated on membrane	Domestic wastewater Food, pharmaceutical and pulp industries	30 min	Suitable for sterilised and artificial samples Uses commercially available instrument	Sample handling Filtration needed (−clogging problem) Temp controlled lab 'Soluble BOD' only	Iranpour *et al*[38]
(B)OD	Microorganisms isolated from activated sludge (immobilised)	Wastewater	6 min (6.6–32.8 mg dm^{-3} OD)	Short response time Not affected variations in wastewater quality Uses realistic mixture of microbes (not just one species)	Influence of temp and pH Storage at 4 °C Limited lifetime and stability	An *et al*[9]
BOD	Luminous bacteria in suspension	Synthetic, municipal and food factory wastewater	15 min 5–120 mg dm^{-3} (linear)	Temp range (18–25 °C) Disposable biosensor Easy storage and reactivation of cells	Single use Difficult for on-line (preparation and maintenance of cultures) Only one species pH sensitive	Hyun *et al*[46]
BOD	Biofilm with preozonation of refractory organic compounds	Artificial wastewater, river waters	~5 min + 20 min (DL:0.2 mg dm^{-3})	Rapidity Sensitivity	Sample pretreatment Chemicals needed pH and temp sensitive	Chee *et al*[77]
N-BOD	Disposable microbial sensor	Ammonium standard solution and municipal wastewater	6–12 min	Specificity Short response time Measures nitrification and inhibition of nitrification Possible automation	Short lifetime (3 days) and stability Storage Long recovery time (45 min to 2 h) and calibration	Konig *et al*[78]
Headspace BOD ($HBOD_5$)	Similar to BOD_5: oxygen demand from liquid phase and container headspace	Municipal wastewater	5 days	Representative of real conditions No dilutions Inexpensive Similar to standard method	Time required, not suitable for on-line monitoring Sample transfer for DO measurements	Logan and Wagenseller[20]
$HBOD_3$	GC-based HBOD test	Municipal wastewater	3 days	Improved HBOD method (no sample transfer) Attractive alternative to BOD_5 test	Time required	Logan and Patnaik[26]
BDOC (Biodegradable Dissolved Organic Carbon)	Biological (inoculum from the sample)	Reclaimed and municipal wastewater	28 days (also 5 days) (4–15 mg dm^{-3} of DOC)	Specificity Insensitive to nitrifiation and inorganic oxidations Good precision and accuracy at low DOC concentrations	Filtration and sample handling Dilutions (for unknown samples) Complicated and time consuming laboratory-based technique	Khan *et al*[10,11]
SBOD (Soluble BOD under certain conditions)						
Bacterial biomass-COD	AODC (Acridine Orange stain Direct Counting of bacteria)	Wastewater and wastewater sludges	Few minutes (5–10)	Specificity Useful for predictive mathematical models Not affected by particulate matter	Specificity Chemicals and sample handling Equipment Time consuming	Munch and Pollard[79]
Readily biodegradable COD (RBCOD)	Single-OUR method	Wastewater	Few tens of minutes	Specificity Useful information for mathematical models	Specificity Calibration and dilutions required for high COD samples Temp sensitive	Xu and Hasselblad[80]

expressed the need for more complex BOD check solutions. These would preferably contain solutes with different molecular sizes and structures commonly found in wastewater.

Biosensor measurements usually take a few minutes or hours. While this can be an advantage over conventional methods, less reactive substances of low diffusivity and/or enzymatic oxidation rates may be neglected during this short period.[47] Because of the biochemical nature of the oxidation reaction involved, temperature and pH are limiting factors that must be controlled accurately (Figs 3 and 4). Live cells also demand particular storage conditions and feeding when not in use, whereas the re-activation of dead cells, which allow easier and larger storage (and commercialisation), can be time consuming and take up to a few days. On the other hand, Osbild and Vasseur[40] remarked that high concentrations of nutrients may also have adverse effects and decrease sensitivity. Finally, biosensors have a short lifetime, from a few days to a few months, which limits their application to continuous on-line monitoring.

Existing measuring systems can be classified into batch-type and flow through, or flow injection type devices.[49] Most commercially available BOD sensors are flow-type systems that can be more easily automated, but generally require high maintenance to prevent fouling and clogging. In a recent study, Osbild and Vasseur[40] presented a detailed review of a few major types of biosensors, along with their characteristics, and also discussed the needs of biosensor technology development. Respirometric type biosensors mainly rely on the assumption that oxygen demand is proportional to the organic content of the wastewater being analysed.

Alternative techniques have recently emerged and

Table 3. Current techniques in monitoring wastewater organic load: optical sensors and non specific sensor arrays

Parameter	*Description*	*Applications*	*Range and response time*	*Interests*	*Limitations*	*References*
BOD	Non-specific sensor array (electronic nose)	Domestic wastewater	Less than 10 min	Rapidity Non-invasive No reagents Versatile technique Good potential for on-line monitoring Uses commercially available instrument	Relationship is source/site specific and time dependent Further development needed	Stuetz *et al.* [72,73]
BOD COD	Oxidation by hydrogen peroxide with UV light	Diluted raw sewage, synthetic sewage and food industry effluents	55 min 0–54 mg dm^{-3} (BOD) 25–150 mg dm^{-3} (COD)	Fully automated Can be used for on-line monitoring Uses non toxic reagents	Limited range Range and correlation are source dependent	Guwy *et al.* [8]
BOD COD TOC TSS Nitrates and anionic surfactants	UV spectral measurements and multivariate calibration	Wastewater surface waters	Few minutes 0–250 mg dm^{-3} (BOD) 0–500 mg dm^{-3} (COD) 0–150 mg dm^{-3} (TOC)	Rapiditiy Versatility Good potential for automation and on-line field monitoring of TOC, COD and nitrates Commercialy available	Sample handling Acquisition of reference spectra and calibration necessary for samples of different origin Not as good for BOD	Thomas *et al.* [1,57]
BOD, nitrates, (TOC and COD)	Optical scattering (fluorescence)	Wastewater and surface water	Real-time	Real-time measurement Non-invasive Potential for screening and on-line monitoring	Still in infancy Research needed Fluorescence affected by pH and temp Correlation with BOD is plant site selective	Reynolds[7] Reynolds and Ahmad[58] Ahmad and Reynolds[68]
BOD	UV absorption (280 nm)	Farm wastes	Few minutes 100–10000 mg dm^{-3}	Indicates BOD range and dilution required for true BOD_5 Potential for rapid gross measurement in the field	Poor sensitivity Uses only one wavelength Interferences from particles and toxic metals	Brookman[13]
COD TOC	UV absorption	Municipal wastewater	1 min	No chemicals Relatively inexpensive On-line and real-time Sensitivity	Immerged sensor (fouling) Influence of suspended particulate material	Matsche and Stumwohrer[54]

have proved useful in characterising wastewater quality. Gernaey *et al*[12] and Rozzi *et al.*[50] recently looked at the development of titration biosensors. This relatively new family of instruments measure the activity and the proton consumption, or production rate, of a microbial population in a reactor vessel. In

Table 4. Current techniques in monitoring wastewater organic load: modelling and virtual sensors

Parameter	*Description*	*Applications*	*Range and response time*	*Interests*	*Limitations*	*References*
COD, biomass and nitrogen	Modelling from known percentage of organic fractions	Activated sludge		Could help detect anomalies, or predict changes in complicated bioprocesses?	Still in infancy Lack of experience and basic understanding Better characterisation needed	Henze[76]
BOD	Prediction using artificial neural network	Paper mill effluent	Few hours	Rapid prediction Good accuracy if bias monitored and feedback algorithm applied	Many other parameters needed Rely on operator to enter imput variables Affected by long term changes in wastewater quality Retraining and updating every 6 months	Masmoudi[75]
BOD	Mathematical prediction from COD values	Domestic, poultry and brewery wastewaters	~2 h (COD)	Rapid surrogate measurement	Not a real measurement Knowledge and dataset needed for each particular wastewater Assumes a constant and linear relationships exists all the time	Ademoroti[21]
COD NH_4 NO_3	Artificial neural network + multi sensor (pH, temp conductivity, redox potential DO, turbidity)	Influent (COD, NH_4) Aeration basin (NO_3, NH_4)		On-line Rapid surrogate measurement For monitoring treatment efficiency only	Approx/estimation Training needed Problemb in case of sudden changes in wastewater composition Reliable for a short period only	Hack and Kohne[81]
RQ value (linked to COD/TOC ratio)	Off-gas analysis (CO_2 and O_2)	Wastewater treatment plants		Can be used to measure COD if RQ combined with TOC measurement	Does not distinguish C-oxidation from N-removal Only big changes in nitrification activity can be monitored	Hellinga *et al*[82]

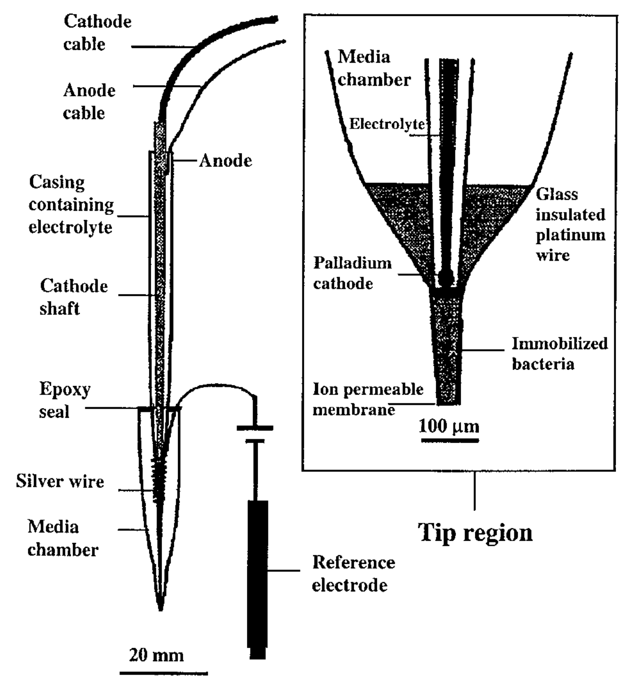

Figure 1. Biosensor based on immobilised bacteria.[4]

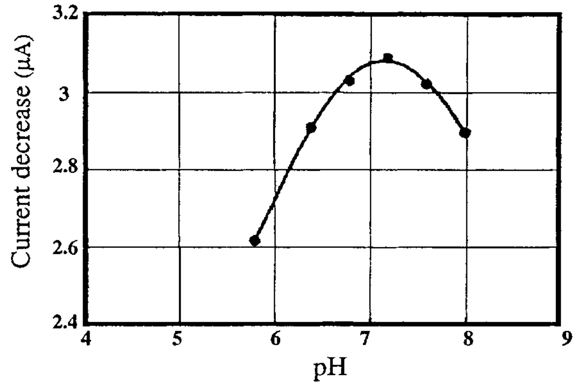

Figure 3. Influence of pH. A GGA standard solution of 13.2 mg dm^{-3} OD was used at 30 °C.[9]

their reviews, the authors also described new advanced developments of hybrid respirometers where titrimetric techniques are coupled with respirometric techniques. Although indirect characterisation of the biodegradation of readily biodegradable organic carbon (rbCOD) have been investigated,[51] these new technologies are generally still limited to ammonia and nitrification/denitrification studies (Table 5).

4.2 Optical sensors

Optical techniques have a long history of use for chemical analysis and water quality monitoring. These techniques, based on the interaction of light with the sample, can be classified as light absorption measurements (UV-visible spectrophotometry, IR spectrometry) and fluorescence measurements (spectrofluorometry). The characteristic transmission, absorption, fluorescence spectrum or vibrational properties of a chemical species is measured in order to determine its concentration or identity.[52]

Over the past few decades, the growing need for reliable on-line monitoring resulted in a considerable amount of research and development that has been devoted to more rapid and new techniques. The appearance of a wide range of near on-line instruments has been encouraged by the 'optoelectric revolution' and associated lower costs, as well as the improvement of existing devices, light sources, detectors, and new optical material.

The uses and limitations of these techniques in water quality and bioprocesses monitoring, as well as the potential of some new concepts in sensor technology, have been extensively reviewed and discussed in recent literature.[52,53]

The major advantages of optical techniques lie in their rapidity and versatility. Additionally, their relatively low running costs, the absence of chemicals, and the limited, or absence of sample handling, make them highly desirable for real-time on-line monitoring for a wide range of parameters. Therefore, the range of optical sensors and measurement techniques continues to expand as new optoelectronic devices and material become available, with researchers moving

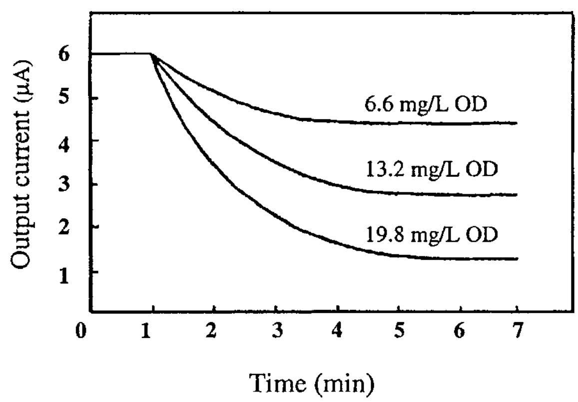

Figure 2. The typical response curves of the OD biosensor for GGA standard solution.[9]

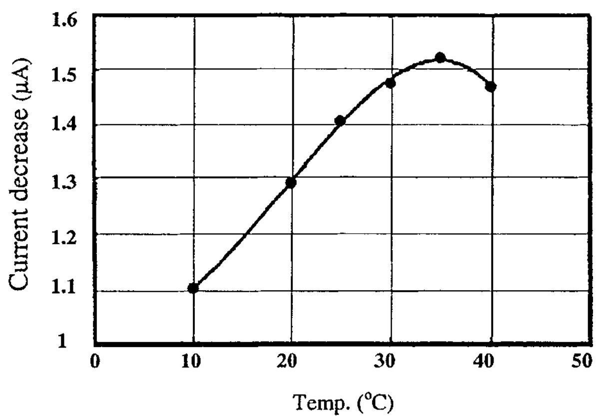

Figure 4. Influence of temperature. A GGA standard solution of 6.6 mg dm^{-3} OD was used at pH 7.2.[9]

Table 5. Investigated applications of titration biosensors (after Rozzi *et al*[50])

Conversion	Type	Substrate	Product	Titrant	Application name
Nitrification	Acidif	NH_4^+	H^+	NaOH	ANITA
Denitrification	Alkal	NO_3^-	OH^-	HCl	DENICON
Denitrification	Alkal	rbCOD	OH^-	HCl	DENICON-BND
Dephenol	Acidif	Phenol	H^+	NaOH	ANITA
Methanogenesis	Alkal	Acetic acid	HCO_3^-	Acetic acid	MAID

towards waveguide-based systems and integrated optics.[52] However a number of problems still need to be overcome, including biofouling of the probe tips, calibration stability and selectivity.[52] For example, light absorption/scattering measurements are often affected by the presence of air bubbles in solutions which can cause interference to the optical signal, and may result in errors. Similarly, agitation and aeration may result in noise in the data, and high concentration of suspended particles can be a major limiting factor.[13,53,54]

Some early studies, based on the knowledge that many pollutants in water and wastewater, and dissolved organic compounds with aromatic structures strongly absorb UV radiation, have used UV spectrophotometry to determine dissolved organic matter in stream water[55] and also to measure total organic carbon in waste water.[56]

In a recent attempt to estimate BOD_5 by using UV spectrophotometry, Brookman[13] investigated the absorbance at 280 nm for slurry and farm effluents. The approach proved useful for rapid estimations of BOD levels and the indication of dilution ranges for standard BOD_5 tests, but it showed poor sensitivity and was affected by interference from particles and toxic metals. Alternatively, Matsche and Stumworher[54] showed the good correlation of UV absorption at 260 (or 254) nm with COD (and TOC), and the interests of additional measurements at a second wavelength (eg 380 nm) to reduce the influence of particulate material. The technique is rapid (1 min), relatively inexpensive and offered good sensitivity without the use of chemicals and sample preparation. The major drawback, however, lies in the use of an immersed sensor for on-line monitoring which is prone to fouling (Fig 5).

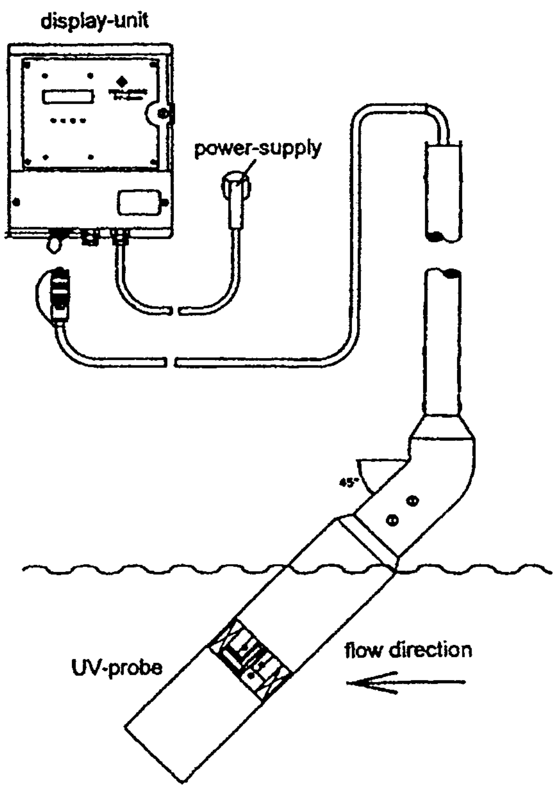

Figure 5. UV-probe.[54]

Thomas *et al.*[57] discussed the significance of UV absorption and showed some of its limitations for the quality control of wastewater when using a small number of wavelengths (one or two). In their studies, the authors demonstrated the potential of UV spectral measurements with a deconvolution method for the estimation and on-line monitoring of specific parameters such as TOC, COD and BOD (as well as nitrogenous and phosphorous compounds by the addition of chemical reagent). This rapid and versatile technique is now commercially available.[57] Despite the fact that this is a flexible and well developed technology, some inadequacies still remain, associated with sample handling and the need for the acquisition of reference spectra. Finally, not all compounds (eg carbohydrates, saturated hydrocarbons) absorb at the specified wavelength and thus cannot be considered with UV spectrophotometry. This may lead to uncertainty in the estimation of factors such as BOD.

Although the technologies discussed above have gone some way to enhance monitoring, they continue

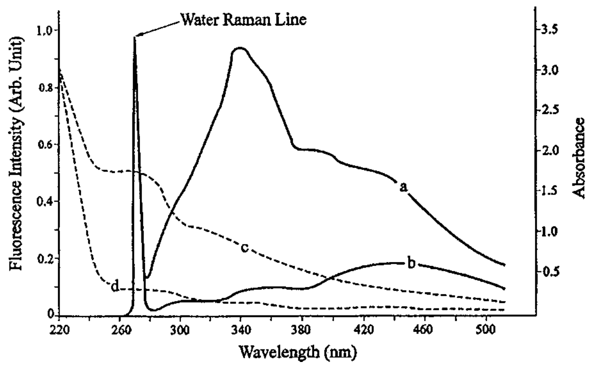

Figure 6. Typical fluorescence spectra of: (a) settled sewage; (b) treated effluent, and absorption spectra of: (c) settled sewage, (d) treated effluent.[61]

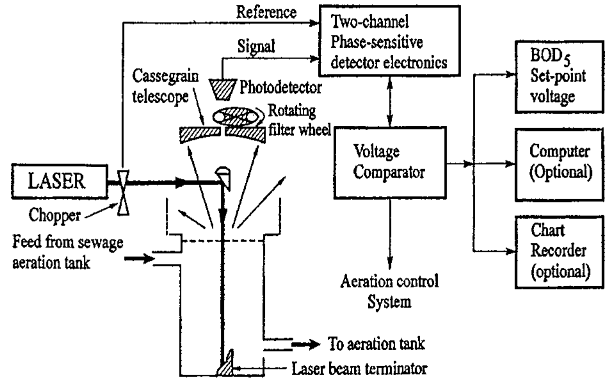

Figure 7. Scheme for non-invasive continuous monitoring of water quality for process control in water treatment plants, based on the detection upwelled fluorescence.[61]

to be invasive and are often not robust enough for on-line monitoring applications.[7] As Reynolds and Ahmad[58] noted, in most cases research was primarily concerned with establishing a correlation between UV absorbance at 254 nm and the dissolved organic matter or carbon in natural waters, with the fouling of optical cell components resulting in a loss of sensitivity, reproducibility, and the need for frequent re-calibration. As a consequence, the use of fluorescence properties of natural waters to determine the presence of organic matter and to measure total organic carbon, or dissolved organic carbon, has been extensively studied in recent years (Fig 6).[59–61]

In particular, Reynolds and Ahmad,[58] Ahmad and Reynolds[61] and Ahmad *et al*,[62] have studied the correlation between BOD_5 values and fluorescence intensity. The authors have shown the feasibility of detecting the scattering and fluorescence properties of surface and waste waters using non-invasive remote-sensing technology (Fig 7). The effects of some environmental parameters and the advantages over current methods (BOD, COD, TOC and absorption at 254 nm) are also discussed.

Although the prospects for non-invasive real-time and on-line monitoring of water and wastewater quality may seem promising and are indeed very attractive, the technology is still in its infancy, and further research and development is required. Specifically, research regarding the effects of environmental factors such as temperature, changes in pH and the concentration of metal species need to be considered.[7]

4.3 Sensor arrays

In the last two decades there has been increasing interest in the development of sensor array technology. The commercialisation of so-called ‘electronic noses’ which are capable of detecting and recognising complex odours started to take in place in the mid-1990s.[63] Although this is a new technology, electronic nose instrumentation has already been used successfully in many fields of applications, such as foodstuffs, beverages, perfumes.[64–68] This demonstration that electronic noses can discriminate between different samples of various nature, as well as monitor their stability and changes in quality with time, was mainly made possible by the recent developments in computing technologies and our better understanding in the field of artificial intelligence.[69]

Consequently, environmental monitoring became one of the most demanding application areas for electronic noses.[63] Given its non-invasive and versatile character, and its potential for real-time and on-line monitoring of water and wastewater quality, the technology acquired growing interest in the water industry. Recent research has demonstrated the ability of electronic noses to detect low levels of organic pollutants and tainting compounds in both waste water and potable water, and that it is possible to distinguish between wastewater samples of different types and origin.[70,71]

Other similar research involving the analysis of real wastewater samples has shown that non-specific sensor arrays can be useful in monitoring organic pollution as a good correlation between the sensors' responses and the 5-day BOD test was observed for time periods of 4 weeks or less.[72,73] The findings showed the enormous potential of the technique, and suggested that it could be used for predicting BOD values (or other organic load parameters), as well as for monitoring and/or controlling the biochemical activities of a wastewater treatment plant. However, this is still a novel technology and much of the original work was carried out with laboratory-based sensor systems, which were developed for other applications such as foodstuffs, beverages and perfumes discrimination. Additionally, most currently available sensors show some sensitivity to changes in temperature, humidity and flow rate.[63] In principle, these effects can be minimised by careful system design and sample handling, but this can make the overall instrument more complex and expensive, and can also limit sample throughput. This remains a challenging problem with one possible approach being to use

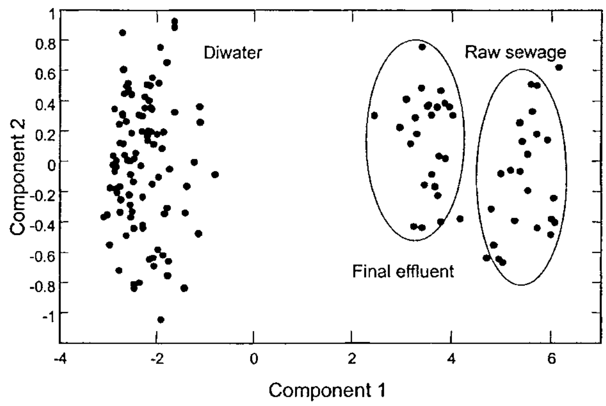

Figure 8. Plot of principal components showing the separation of raw sewage, final effluent and RO water (110, 30 and 30 replicates respectively) using a temperature-controlled flow-cell and a 12 conducting polymer sensor array.[74]

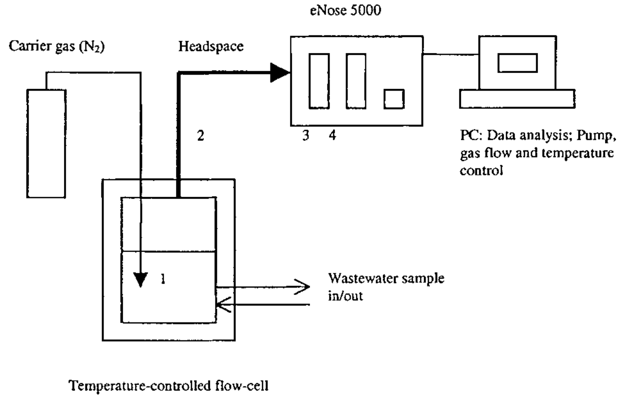

Figure 9. Schematic of experimental apparatus for on-line and non-invasive monitoring of wastewater quality, showing gas sparger (1), temperature-controlled sample transfer line (2), gas flow, humidity and temperature control module (3) and sensor array module (4).

temperature (and/or relative humidity) as an input to artificial neural networks, which afford some parametric compensation.

Bourgeois and Stuetz[74] recently demonstrated the advantages of using a simple (temperature-controlled) external flow cell for generating a reproducible sample headspace and minimising both relative humidity and temperature variations (Fig 8). This sampling technique (Fig 9) is currently being evaluated for continuous wastewater monitoring and preliminary results have shown that the methodology is reproducible and that good predictions of wastewater organic load can be achieved when an appropriate data analysis protocol is used (Bourgeois *et al*, unpublished). With further development, the technology could serve a number of applications and provide a simple technique for real time and non-invasive monitoring of wastewater quality.

4.4 Virtual sensors

Software sensors, or 'virtual sensors', have been defined by Jacobsen and Lynggaard-Jensen[37] as a term used for 'signals' obtained from calculations using measured signals from reliable, available sensors in combination with other signals such as on/off indications and time counters. Such 'sensors' emerged from the need for rapid evaluation/prediction of essential parameters for which data are not readily available. In most cases, the lack of real-time/continuous data is associated with technical limitations or cost restrictions.

Some virtual sensors have been used as filters or as real-time quality controls of datasets and sensor validation procedures. The technique has also proved useful in the field of modelling and the determination of environmental parameters, such as in the case of wastewater treatment plant designs. This allows the rapid prediction of the effect of changes of a number of factors on the variable of interest. One such mathematical tool is the neural network model, which was used for on-line prediction of mill effluent BOD.[75] The method uses artificial intelligence to build non-linear multivariate models based on large amounts of data. These techniques are not based on fundamental physical or chemical, or even biochemical principles and must be combined with practical insight to be effective. Similarly, in an early evaluation of the modelling of activated sludge processes, Henze[76] reported that there are still many problems to be solved with regard to wastewater characterisation for modelling purposes. Indeed there are many difficulties associated with the lack of experience and/or basic understanding.

5 CONCLUSION

Rapid monitoring of wastewater organic load is of major concern to the water industry. At present there does not seem to be a commercially available instrument that satisfies all the requirements for wastewater treatment monitoring, which will allow real-time control of effluent quality in compliance with the latest environmental regulations. Traditional methods are inadequate for providing continuous and real-time monitoring of water quality. For many years the approach to the measurement of global parameters such as BOD, COD and TOC has been off-line, requiring sample collection and retrospective laboratory analysis.

In this review, the limits of the current standard procedures together with the importance of on-line and real-time control for automated systems have been demonstrated. Potentially suitable techniques such as biosensors and optical (absorption) sensors were presented and discussed. Many of these are still typically limited by environmental factors, short lifetime or fouling problems. New innovative and non-invasive techniques using optical (fluorescence) and headspace (sensor arrays) analyses techniques (or even virtual sensors) have also been described. However, these promising new methods are still in their infancy and require further development.

Despite the increasing range and diversity of techniques and technologies available for wastewater monitoring, there are still a number of barriers to be overcome before they are accepted and implemented by the wastewater treatment operators. For instance, improvement in comparability and reliability of measurements is clearly needed. Similarly, clear definition and standardised validation of water quality determinants is also essential. Finally, a system's success and commercial viability will be determined by its robustness, simplicity, rapidity and cost considerations.

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